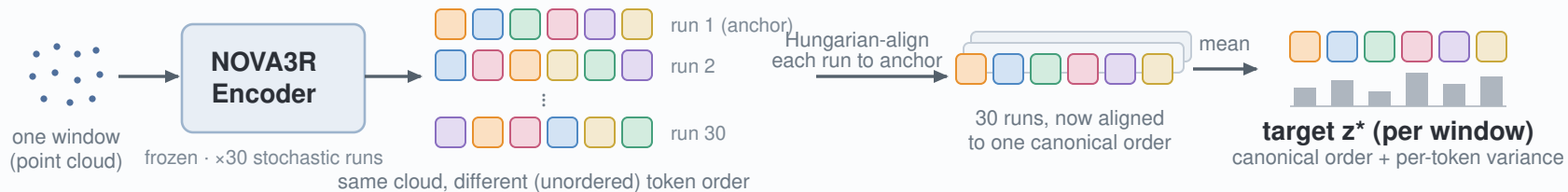
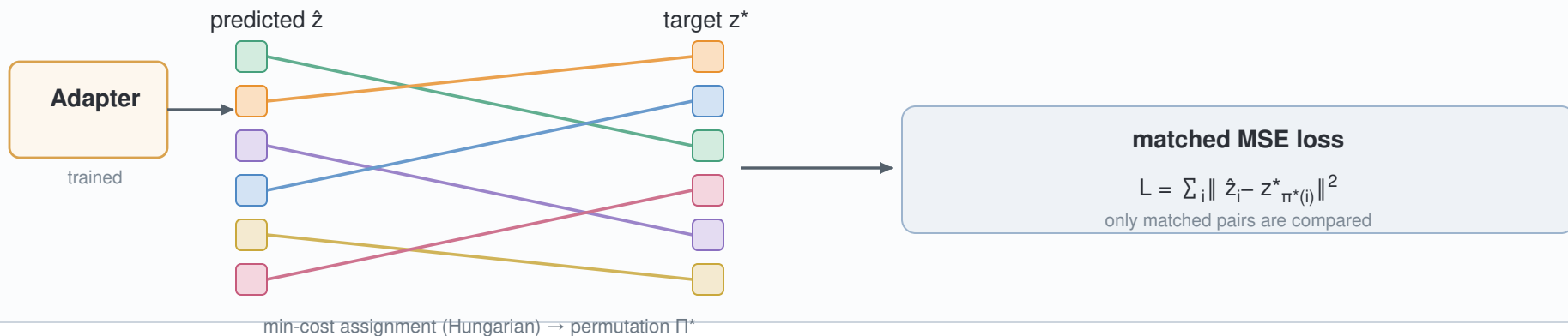


1 Building the target order (offline — done once per window)



2 Hungarian matching in the training loss (every step)



Tokens are a permutation-invariant **set** — matching runs **within a single window** only
samples $\leftrightarrow$ anchor when building z^* , and $\hat{z} \leftrightarrow z^*$ when training — never across different scenes / point clouds